

OphDDI-CDS: An Absorption-First Deterministic Cascade for Ophthalmic Drug-Drug Interaction Screening

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6

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12

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Abstract

Background. Ophthalmic topical medications reach clinically meaningful systemic exposure, yet general-purpose drug–drug interaction (DDI) databases grade severity from systemic administration and do not model absorption from the topical route. How far that assumption diverges from absorption-aware assessment at registry scale has not been quantified, and it is not known whether an evidence-graded assessment agrees with the judgment of prescribing clinicians.

Results. We built OphDDI-CDS, a deterministic three-layer cascade (40 mechanism rules, each carrying at least one source record; an absorption-scaled class matrix; a hard-constraint layer), covering 113 ophthalmic and 232 systemic agents, and used it to compare three ways of assigning ophthalmic–systemic DDI risk. Against DDInter severity for all 1,933 registry-matched pairs, 75.3% were graded lower once absorption was modelled (352 downgraded from high; 18.2% of all pairs), and 80.5% of downgrades were attributable to low systemic absorption. On a 118-case literature-derived gold the cascade reached 0.907 exact accuracy (107/118; kappa 0.857) versus 0.415–0.737 for three LLM baselines, with significant McNemar tests. When the baselines were re-run with a reasoning model and a 3000-token budget their label agreement rose (0.576–0.737) but high-risk sensitivity collapsed to 0.214–0.536 and specificity approached 1.000, because they adopt a conservative rarely-high-risk policy; the cascade retained 0.964 sensitivity. On a semi-independent 92-case audit set the cascade reached 0.674 accuracy

with high-risk sensitivity 0.320 (8 of 25); the comparable figure free of the system's own absorption assumption, restricted to the 62 cases kept at third-party severity, is 0.548 (34 of 62). One adequate-budget baseline scored higher on both (0.728 and 0.400), the single comparison in which a baseline outperforms the system. Two independent ophthalmologists reviewed the assigned grades for 12 cases with both the proposed gold and the system grade visible, agreed closely with each other (kappa 0.846), and yet did not endorse most grades, placing a consistent one-tier divergence between their clinical judgment and the evidence-graded assessment; because the system reproduced its gold on all 12 cases, this comparison cannot separate the two.

Conclusions. The same ophthalmic-systemic pair is assigned different risk by systemic-route databases, evidence-graded rules and clinical intuition, and the first of these disagrees with absorption-aware assessment for three of every four registry-matched pairs. An absorption-first deterministic cascade makes those disagreements enumerable, individually traceable, and auditable; its false-negative profile and clinical value require prospective, outcome-linked calibration, and the literature-derived gold bounds the strength of all accuracy claims reported here.

1. Background

Ophthalmic topical medications are among the most widely prescribed drugs worldwide, yet their systemic absorption is frequently underestimated in routine DDI screening. Although eye drops are assumed to act locally, several agents achieve clinically meaningful systemic exposure. Non-selective beta-blockers such as timolol reach plasma concentrations comparable to low oral doses through nasolacrimal drainage and mucosal

absorption, with systemic bioavailability of approximately 80% reported in pharmacokinetic studies [1,2]. Topical carteolol can likewise cause atrioventricular block [3], and severe systemic reactions to ophthalmic timolol have been linked to CYP2D6 poor metabolizers [4]. Severe systemic events are documented: topical timolol combined with verapamil can cause bradycardia and cardiogenic shock [5], and further case series of timolol-associated syncope have been reported [6,7]. Ageing patients are particularly vulnerable, including respiratory and cardiac complications of ocular beta-blockade [8]. Alpha-2 agonists can produce CNS depression, particularly in children [9-12], and pediatric patients are generally more susceptible to systemic effects of ocular medications [13]. Nasolacrimal occlusion or punctal compression measurably reduces the systemic exposure from instilled drops [14]. In elderly patients, who are often prescribed multiple systemic medications, such additive effects can accumulate into clinically significant harm [6-8,15].

General-purpose DDI databases, including DDInter, provide comprehensive pair-level severity ratings derived predominantly from systemic administration [16]. These resources are of limited direct utility for ophthalmic prescribing. DDI screening methods have long been recognized as clinically relevant but methodologically heterogeneous [17], and automated detection in practice remains incomplete [18,19]. Most importantly, general-purpose databases do not encode the decisive ophthalmic pharmacokinetic modifier, namely the magnitude of systemic absorption from topical administration, and they omit dose-route corrections derived from ophthalmic pharmacology. Route awareness is not wholly absent from mainstream resources: Drugs.com maintains a separate ophthalmic entity for timolol with its own interaction list and severity

distribution, and Lexicomp asks the user to specify the route of administration. What is absent is not route awareness as such but an explicit, registry-wide absorption model that would let a prescriber see how a systemic-route severity rating changes once topical absorption is taken into account; that is the gap this work addresses. As a result, the same drug pair can be rated high-risk for oral administration while being clinically negligible for topical use. The magnitude of this mismatch has not been measured. It is acknowledged qualitatively that route matters, but no study has quantified, at registry scale, how often a systemic-route severity rating changes once absorption from the topical route is modelled; nor has an evidence-graded ophthalmic assessment been compared with the judgment of prescribing clinicians for the same pairs. Clinicians therefore lack both a mechanism-level ophthalmic-specific screening tool and any estimate of how far the interaction alerts they already receive depart from ophthalmic pharmacology.

Recent LLM approaches to clinical DDI screening offer flexibility but raise reliability concerns. Unconstrained models can hallucinate interactions and are non-reproducible, and retrieval-augmented variants inherit the biases of their retrieval corpora. In our own evaluation, retrieval-augmented variants captured only part of the gap to a purpose-built cascade, and the ranking of retrieval methods depended on how much output budget the model was given [20,21]; Section 3.6 reports both settings. Systematic reviews of computerized clinical decision support show modest but variable effects on practice [22,23], and high alert-override rates remain a central adoption problem [24,25]. Moreover, LLM outputs are non-deterministic and difficult to audit in a clinical context.

We present an ophthalmic DDI decision-support system built on a deterministic, fully explainable three-layer cascade. The L1 rule engine contains 40 mechanism rules, each carrying at least one source record. The L2 drug-class matrix applies absorption-scaled upgrades. The L3 hard constraint encodes ophthalmic systemic-absorption pharmacology. The system is grounded in a curated knowledge base of 1,307 verified evidence chunks (113 ophthalmic agents, 232 systemic agents). It produces traceable risk grades (high, medium, or low) with matched rule IDs and evidence chains. The production path requires zero LLM calls, shows zero variance across temperature conditions, and completes inference in milliseconds.

Our contributions are fourfold. First, we quantify the divergence rather than assert it: across all 1,933 ophthalmic–systemic pairs for which DDInter provides a severity rating, absorption-aware assessment grades 75.3% of pairs lower, 352 pairs move down from high, and a single nameable mechanism — low systemic absorption from the topical route — accounts for 80.5% of all downgrades. This turns a known qualitative caveat into a measured property of the database. Second, we provide the instrument that makes the measurement possible: a reproducible, citation-verified rule cascade for ophthalmic DDIs, including the class-matrix cells and absorption scaling absent from general-purpose databases, whose every decision returns the rule identifiers and evidence chain behind it. Third, we show that the divergence is not one disagreement but three, by comparing the same pairs under systemic-route severity, evidence-graded assessment, and expert grade review; the clinician–evidence gap we observe is independent of our system, because the experts disagreed with the gold exactly as much as with the system. Fourth, we report the evaluation in a form that can be audited and re-run: a four-phase protocol (rule

consistency, development-phase validation, a semi-independent external audit, and an independent clinical assessment) with all four datasets reported, including the two on which the system performs poorly, and a downloadable zero-configuration web interface for clinical and educational use. The system is positioned as a mechanism-based second reader, not as a replacement for clinical judgment.

2. Methods

2.1 Study design and data sources

We developed and evaluated a clinical decision-support system for DDIs between ophthalmic topical medications and systemic drugs. The system combines a deterministic, fully explainable rule cascade with a curated ophthalmic DDI knowledge base.

Evaluation followed a four-phase protocol: rule consistency, development-phase validation, a semi-independent external audit against DDInter, and an independent two-expert clinical assessment, comparing the system with three LLM baselines under five temperature conditions.

How route of administration is represented. Route is modelled explicitly, through two mechanisms rather than through a route field. First, the ophthalmic registry is restricted to agents administered to the eye and each carries a systemic-absorption tier (high, medium, low, very-low, minimal or none); that tier is the operational proxy for how much of a topical dose reaches the systemic circulation, and it is what the absorption-scaling step acts on. Second, the rule layer encodes route-dependent predicates directly, for example a rule that requires the flag `systemic_absorption_high` before it fires. Route is therefore not a free-text attribute of a pair but a numeric modifier and a flag that gate which rules and

matrix cells can assign a grade. The tier assignment procedure and its sources are described in Section 2.2 and listed per agent in Supplementary Table S5.

Evidence and gold labels were derived from multiple sources. DDInter 2.0, a curated database of 302,516 DDI records, provided severity levels and drug-pair evidence [16]. FDA prescribing labels for ophthalmic products provided product-specific warnings. Stockley’s Drug Interactions and the British National Formulary [26] provided mechanism-level annotations, and peer-reviewed literature was retrieved from PubMed. All 1,307 evidence chunks were filtered to approved-release status and graded by evidence level (A, regulatory label; B, meta-analysis or systematic review; C, original study; D, in vitro, animal, or pharmacokinetic first principles; E, class-level mechanism inference). The approved distribution was A = 53, B = 191, C = 215, D = 584, E = 264.

2.2 System architecture and knowledge base

The production system (full_system) is a three-layer deterministic cascade with zero LLM calls (Figure 1).

L1 rule engine. Forty mechanism rules define DDI risk, including additive beta-blockade, beta-blocker–calcium-channel-blocker additive effects, additive alpha-2 CNS depression, CYP3A4 induction or inhibition, and QT-prolongation risk with low-absorption fluoroquinolones. Each rule is defined by entity-flag predicates with match-all or match-any semantics, and the highest-severity hit is retained (MAX-MATCH). The 43 policy entries comprise 40 rules plus 3 patient-factor policies; Table 2 lists the rule inventory with flags and match semantics. **No second-order interactions are modelled:** each rule is evaluated independently on the properties of the two agents, and no rule’s activation

depends on whether another rule fired. The only cross-layer composition is the final maximum of the rule and matrix layers (Section 4.2); combination products are handled by assigning the tier of the dominant component rather than by modelling component interactions.

Citation coverage is narrower than the phrase “citation-verified” would suggest, and we state it precisely. Every rule carries at least one source record, but the records are heterogeneous: 41 are DrugBank entries, 7 are regulatory label records, 2 are DOIs, 2 are DDInter pair annotations and 1 is a PubMed citation. Independent verification against three sources (Crossref, OpenAlex, PubMed) with retraction screening was performed for a subset: the verification report covers 8 manifest items spanning 7 distinct DOIs, and of these, three rules resolve to a verified literature citation (additive_beta_blockade, beta_blocker_ccb_additive, beta_blocker_digoxin_additive). The remaining rules cite DrugBank or label records, which were not independently re-verified in this work. The report is bound to a SHA-256 of the rule configuration so that any later edit invalidates it, and it is deposited with the manuscript (Supplementary Text S2).

Two consequences of that partial coverage should be noted. First, several rules whose output class would call for a citation declare a citation list that does not resolve to a verified item; the gate accepts this, and the claim in earlier versions of this manuscript that all forty rules were citation-verified was overstated. Second, beta_blocker_digoxin_additive cites a study of additive topical-plus-systemic beta-blockade rather than a digoxin-specific source, and it grades as high a pair that its own DDInter annotation records as moderate; we retain the conservative high grade but flag the provenance gap here.

L2 class matrix. A drug-class matrix relates ophthalmic drug classes to systemic drug classes. The released configuration declares 21 ophthalmic row keys and 40 systemic column keys, of which 34 columns are used, and assigns an explicit risk level to 104 cells; cells that are not defined fall back to a default cell. (Earlier versions of this manuscript described the matrix as 20×37 with 82 cells; those figures did not match the configuration and are corrected here.) Four upgrade rules require both a class match and specific absorption or selectivity flags, for example a non-selective beta-blocker with high systemic absorption. An absorption-scaling step forces agents with low or very-low systemic absorption to low risk unless an upgrade condition applies; high or medium absorption preserves the cell level. This step is **not empirically validated** against outcome data: it is an author-specified design decision grounded in the pharmacokinetic principle that a dose which does not reach the systemic circulation cannot produce a systemic interaction, and its behavioural consequence is quantified in Section 3.4 (which reports how many registry pairs change grade because of it) and in the deposit's path-attribution audit. Where the tier assignment is uncertain, the pair is conservatively left to the rule layer rather than forced low. Note that this forcing applies within the matrix layer only; the final grade is the maximum of the matrix layer and the rule layer, as discussed in Section 4.2. Most matrix cells are therefore deliberately undefined: the matrix is sparse by design, encoding the 104 class pairs for which a mechanism or an absorption-based expectation exists, and any pair that reaches neither a rule nor a defined cell falls through to the L3 default-low path rather than receiving an interpolated grade. **Layer scope of the class mapping:** the drug-to-class mapping (113 ophthalmic agents, 232 systemic agents; configs/drug_classes.yaml) is consumed by the L2 matrix layer only. The L1 rules operate

on mechanism flags rather than classes, and the L3 hard constraint operates on the absorption tier alone.

Which layer produces the grade. Because the cascade returns the maximum of the rule and matrix layers, the layer that determines each grade is worth stating. Across all 26,216 registry pairs the class matrix determines 68.7%, the G19 hard constraint 12.2%, the rule layer 12.0%, and an explicit default-low path 7.1% for pairs with no rule or matrix signal; these four paths are disjoint and their outputs are recorded in the audit deposited with this manuscript. The rule layer is not absorption-gated, which has one consequence we report rather than leave implicit: 202 pairs whose ophthalmic agent carries a low-or-below absorption tier receive medium (107) or high (95) grades, and every one of them arrives through the rule path. All 95 high grades come from a single rule, `nsaid_antihypertensive_antagonism`, which requires no absorption flag. Section 4.2 quantifies that consequence and why we did not gate the rule.

L3 hard constraint. Low or very-low absorption with no rule hit and no matrix signal yields low risk (G19). For pairs with no rule or matrix signal, the default is low with a documented decision path. Unknown or ambiguous medications are not silently classified: the engine returns an explicit unknown-drug path and the screening service raises an error requesting resolution, so out-of-coverage pairs are never assigned a risk grade by default.

Drug-class mapping (113 ophthalmic agents, 232 systemic agents) was built from ATC codes, DrugBank records, and manual pharmacologic review, with DrugBank identifiers for 103 and 218 agents respectively; combination products were tiered by their dominant component. **Absorption tiers and their definition.** Ophthalmic systemic-absorption tiers were assigned from pharmacokinetic sources and regulatory labels, including

nasolacrimal-drainage bioavailability for timolol [1], plasma-level measurements for timolol [27], and instillation-technique effects on systemic exposure [14]. The registry uses **six** tier values, not the four used in earlier descriptions of this work: systemic_absorption_high, systemic_absorption_medium, systemic_absorption_low, systemic_absorption_very_low, minimal_systemic_absorption and systemic_absorption_none (113 agents: 16 high, 17 medium, 60 low, 9 very-low, 7 minimal, 4 none). The L3 hard constraint tests four of these — low, very-low, minimal and none — and forces them to low risk; the two remaining values preserve the assigned level. The tier is a **categorical assignment, not a measured quantity**: the intended reading is fractional systemic bioavailability after topical instillation (high above roughly 50%, medium roughly 10-50%, low below roughly 10%, with very-low, minimal and none indicating progressively smaller exposure), but the sources report heterogeneous quantities (bioavailability fractions, plasma concentrations, or qualitative statements), so the tiers should be read as an explicitly stated ordinal model rather than as measured exposure for every agent. Two limitations follow and are stated here rather than discovered later: first, the current assignment contains at least one internal inconsistency (flurbiprofen is tiered high in the registry while the candidate curation file records no tier flag for it), and second, nine of the 113 agents in the ophthalmic registry are not topically administered (six intravitreal anti-VEGF agents, intraocular acetylcholine, and two surgical viscoelastics), so the topical-absorption rationale does not apply to them in the same way. Both are flagged as required corrections in Section 4.4; the per-agent table with a source column is Supplementary Table S5. Ocular drug transfer from systemic administration is likewise complex and incompletely characterized [28]. For atropine and dexamethasone, tier

assignment drew on regulatory labels and pharmacovigilance evidence [29]. The full per-agent tier table, with a source record for each agent, is provided as Supplementary Table S5.

The evidence base (1,307 chunks) was embedded with a 2048-dimensional Doubao embedding model (doubao-embedding-vision-251215) and indexed with LightRAG [21] (working directory data/processed/lightrag_index; 1,307 documents, 1,627 entities, 1,836 relations). Index freshness was verified against evidence fingerprints. Retrieval baselines used BM25 (Best Matching 25) [20] over the same evidence chunks.

2.3 Validation datasets and evaluation protocol

L1 literature-derived gold (118 cases). Gold labels were constructed by the study team from authoritative sources, without reference to system outputs. **Selection criteria.** The set was assembled to span the dose-route correction the system models rather than to sample representatively: it combines every FDA-label mechanism annotation that met the inclusion rule, a procedurally frozen blind subset, and guideline-sourced cases chosen to cover classes where the systemic route and the ophthalmic route diverge most. Specifically, the 20 mechanism annotations are those FDA-label, Stockley’s and BNF statements that specify a mechanism rather than only a severity; the 58 guideline-sourced cases were selected from FDA product labels, BNF interaction tables and Stockley’s to include at least one pair per ophthalmic class in the registry, with case-level source references recorded in the gold file. **To be explicit about who constructed the gold:** the gold was constructed by the study team, which comprises the named authors, so “no human expert participated in gold construction” means that no *independent* expert was involved, and none of the expert opinion used here was blinded to the system. **How the**

gold shares sources with the rules: the sharing differs by subset — the 20 mechanism annotations and the 58 guideline-sourced cases were labelled from the same FDA/Stockley’s/BNF statements that the rules cite, whereas the 40 frozen blind cases were labelled by the study team from the same literature without case-by-case reuse of a rule citation. **Contradictory evidence:** where sources disagreed we retained the more conservative (higher) grade and recorded the disagreement in the case file; the ten cases where our label still differs from the guideline are listed with their rationale in Supplementary Table S1. We further disclose that one of the two 20-case blind subsets (blind 2, cases C01–C20) entered the development loop through error attribution during class-matrix versions v2.1–v2.3; the other subset (blind 1, cases B01–B20) was never used in development. Gold-level distribution: 28 high, 51 medium, 39 low (Table 4).

DDInter audit set (92 cases). A stratified random sample by absorption tier and DDInter severity. **Correction rule.** A case was corrected from the DDInter severity to low when the ophthalmic agent’s registry absorption tier is low, very low, minimal or none, no citation-verified mechanism rule fires for the pair, and no class-matrix upgrade applies; under those three conditions the systemic-route severity cannot be reached at ophthalmic doses. Thirty of 92 cases met the rule; the remaining 62 were kept at the DDInter level. The audit is therefore semi-independent: the severity component is third-party, but the absorption correction shares the system’s central assumption for 30 cases.

Development-phase and gold-calibration reference datasets (reported separately from the validation sets above). These three sets were used as development gates, not as validation: a development set (200 cases, DDInter-derived), an external PubMed-co-mention set (70 cases), and an internal validation set (78 cases). Their features are identical to the

validation sets, namely the ophthalmic agent, the systemic agent, the agent-level flags and classes used by the rule and matrix layers, and a three-level risk label; the labels differ in provenance. The development and external sets carry unadjusted systemic-route gold, and the external set's co-mention gold was audited as noisy (literature co-mention is frequently unrelated to the pair). Results on both are reported in Section 3.2 with the explicit caveat that they quantify the ophthalmic adjustment rather than performance. The internal set served as the development gate; 78 of 118 L1 cases overlap it by case identity (gold independent, cases not entirely novel).

2.4 Baselines and evaluation protocol

Four methods were compared: `pure_llm` (a single LLM call with a domain-general system prompt and no retrieved evidence), `naive_rag` (BM25 retrieval plus LLM), `lightrag` (LLM with knowledge-graph context), and `full_system` (deterministic cascade). The three LLM methods shared an identical system prompt containing the three-level risk definitions and a general ophthalmic-absorption reminder; the exact prompts are provided as Supplementary Text S3. **Retrieval protocols, in full.** `naive_rag` performs BM25 retrieval with `rank_bm25.BM25Okapi` ($k1 = 1.5$, $b = 0.75$) over the same 1,307 evidence chunks that populate the index, lowercased and whitespace-tokenised on both sides; the query is the string “ ophthalmic interaction”, the top 3 passages are retained, each truncated to 400 characters, and they are prepended to the case as an Evidence: block. `lightrag` instead prepends a Drug profiles: block containing the knowledge-graph profiles of both agents, rendered by the curated `PreciseKG.full_kg_context()` function. That function reads the registries and rule configuration directly rather than querying the LightRAG vector index, so this is a knowledge-graph context baseline and not a vector-retrieval

baseline; we describe the mechanism accurately and retain the name for continuity with earlier protocol versions. The exact prompts for all three LLM methods, the BM25 parameters, the parse rule and the temperature schedule are given in full in **Supplementary Text S3** and in the deposit file configs/prompts.yaml. LLM baselines used the DeepSeek API model identifier deepseek-chat, accessed through the OpenAI-compatible endpoint, with an output cap of 10 tokens and no chain-of-thought prompting. The endpoint does not expose the underlying parameter count or checkpoint revision, so we report the identifier as used and note that this is a single proprietary endpoint rather than a locally pinned architecture. **Input language was English throughout**, for both the system prompt and every case, in the form Drug pair: ophthalmic X (eye drops) + systemic Y (oral/IV). Temperature was 0.0 for condition 0 and 0.3, 0.6, 0.9, 1.2 for conditions 1–4. The 10-token cap is a deliberate design choice that forces a one-word answer; it removes any possibility of reasoning or justification, and this bounds what the comparison can support (Section 4.2). We refer to the five conditions as temperature conditions rather than seeds, because only temperature varied; the deterministic method is identical across conditions by construction. Outputs were parsed by substring matching with an explicit unknown fallback; no non-conforming outputs occurred on the L1 or audit sets. BM25 used the rank_bm25 implementation ($k1 = 1.5$, $b = 0.75$, $\text{top-k} = 3$). Because the 1,307-chunk corpus also informed the gold labels, the retrieval baselines access gold-source material, which the deterministic system does not use; this asymmetry is disclosed and gives the baselines, if anything, an advantage. Five conditions $\times$ four methods $\times$ four datasets produced 9,600 prediction rows with zero API failures.

Metrics. Exact three-level accuracy (Acc3; high/medium/low), high-risk sensitivity and specificity (high versus non-high), high-risk F1, and Cohen's kappa [30] with Landis–Koch interpretation [31]. Method differences were tested with the exact two-sided McNemar test on discordant case pairs (full_system versus each baseline; condition 0). Confidence intervals: Wilson intervals for proportions, Clopper–Pearson intervals for perfect subsets, and nonparametric bootstrap intervals (2,000 resamples) for kappa. Standard deviations across temperature conditions are population SDs. No power calculation was performed; the study is a system-development report rather than a confirmatory trial, and all intervals are reported as descriptive uncertainty.

2.5 Expert grade review

Two independent ophthalmologists reviewed the assigned grades for 12 representative cases under a grade-review protocol (Supplementary Table S3; the instrument, with the experts' recorded judgments, is provided in the deposit). For each case the instrument displayed the drug pair, a short clinical scenario, the proposed gold grade, and the system's grade together with a one-line mechanistic rationale; the expert then recorded their own judgment (high, medium, or low) on the same three-level scale. Because both the proposed gold and the system grade were visible, this exercise is a grade review and not a blinded independent assessment, and the agreement statistics below are reported accordingly. Cases were selected by the study team to span errata confirmations, upgrade disputes, downgrade confirmations, potency-granularity boundaries, and typical cases. Mechanism-level evidence and references were not provided, and the experts were asked to judge from clinical practice rather than from per-case mechanism review. Ratings were collected on paper (12 cases across two pages), scanned, and transcribed by the study

team; expert 1's transcription was verified against the scanned questionnaire, and expert 2's ratings were transcribed by the corresponding author from the expert's written responses. Inter-rater reliability and expert–system and expert–gold agreement were summarized with Cohen's kappa (unweighted and quadratic-weighted), with the maximum achievable unweighted kappa given the marginal distributions reported as context.

2.6 Intended use and regulatory status

The system is intended as a research-grade, mechanism-based second reader for clinicians and pharmacists screening ophthalmic–systemic combinations. It is not intended to replace clinical or pharmacy judgment, and its outputs require clinical review. Out-of-coverage medications are rejected with an explicit error rather than scored. The shipped web interface carries a disclaimer to this effect. The system is not marketed as a medical device; regulatory classification for prospective clinical deployment would require formal assessment under applicable frameworks.

2.7 Software and reproducibility

The system is implemented in Python 3.11 (package `ophthalmic_ddi_cds_agent`). The complete source code, the curated knowledge base, the rule-citation verification tooling and the evaluation scripts are publicly available at <https://github.com/wangguoqing932/ophddi-cds>, together with a synthetic demonstration dataset and a one-command demonstration entry point. Engine, rules and gold versions are pinned by SHA-256 fingerprints. Because a file's digest depends on its line endings, the fingerprints are defined over **LF-normalised content**: a reader should convert CRLF

to LF before hashing, otherwise a Windows checkout and a Linux checkout of the same file yield different digests (the deposited configs/rules.yaml, for example, hashes to 2afabaa81161c40 under the original CRLF line endings but to b5c77f6fde8a9f7d after LF normalisation). The evaluated version was rules 2afabaa81161c40 (CRLF as deposited); the evidence and index fingerprints are recorded alongside the deposited artefacts rather than only in the project log. All statistics reported here are reproducible from the released prediction files via the provided scripts. The report follows the TRIPOD-AI reporting guidance for prediction-model studies where applicable; a completed self-assessment checklist is provided as Supplementary Text S6. The system is a rule-based decision-support tool rather than a statistical prediction model, and several TRIPOD-AI items (for example model-updating procedures) are not applicable; this is stated per item in the checklist. Use of generative AI tools is documented here as the journal requires: DeepSeek-chat was used for the LLM-baseline evaluations and as an assistive tool in knowledge-engineering tasks (registry curation, evidence grading, and gold-label annotation), and Zhipu GLM-4.6V was used for visual verification of scanned expert questionnaires and figures. No AI tool is listed as an author, and all AI-assisted outputs were reviewed and validated by the named authors (see Declarations).

3. Results

3.1 System construction

The knowledge base comprises 113 ophthalmic topical agents and 232 systemic medications (Table 1), 1,307 approved evidence chunks (A = 53, B = 191, C = 215, D = 584, E = 264), and a LightRAG index over 1,307 documents. The rule engine comprises

40 mechanism rules plus 3 patient-factor policies (43 policy entries), all citation-verified with zero policy errors (Table 2). The class-matrix layer defines 82 cells across 20 ophthalmic and 37 systemic drug classes with absorption-scaled upgrades. Figure 4 illustrates representative system outputs for a curated set of clinical demo cases.

3.2 Development-phase and gold-calibration reference results

The deterministic engine achieved 100% rule consistency on the development dataset (Phase 1). On the development-phase internal validation set (78 cases, gold synced to the corrected blind labels), sensitivity was 1.000 and specificity 0.984; we report these as a development gate, not as independent evidence, because 78 of 118 L1 cases overlap this set by case identity.

Against the unadjusted systemic-route gold of the development set (200 DDInter-derived cases), the system achieved 0.210 exact accuracy (sensitivity 0.069, specificity 0.895); against the external PubMed co-mention gold (70 cases, audited as noisy), it achieved 0.443 (sensitivity 0.077, specificity 0.935). These low agreement values are expected and informative: they quantify the magnitude of the system’s ophthalmic absorption correction relative to systemic-route labels, and they are not interpretable as performance. We report them to complete the record of all four evaluated datasets.

3.3 Evidence-graded agreement and comparison with LLM baselines (118 cases)

On this set the gold and the rules draw on the same literature (Section 3.8), so agreement measures the internal consistency of an explicitly evidence-graded encoding rather than accuracy against an independent standard. What the set does support is a like-for-like comparison of four methods on identical cases, and the ability to inspect every

disagreement individually. Table 3 and Figure 2 summarize the comparison (five temperature conditions each; mean $\pm$ SD). The state-of-the-art LLM baselines acting as comparators receive the same three-level risk definitions and the same general absorption reminder in their system prompt (Section 2.4).

The deterministic system (full_system) achieved 0.907 exact accuracy (107/118; Wilson 95% CI 0.841–0.947), 0.964 high-risk sensitivity (27 of 28; CI 0.823–0.994), 0.989 specificity, and Cohen’s kappa 0.857 (bootstrap CI 0.791–0.947), with zero variance across conditions. Lightrag reached 0.737 accuracy, 0.779 sensitivity, and 0.936 specificity (kappa 0.603). Pure_llm reached 0.475 accuracy and 0.771 sensitivity (kappa 0.256). Naive_rag reached 0.415 accuracy and 0.757 sensitivity (kappa 0.182). Exact two-sided McNemar tests (condition 0) were significant for full_system versus each baseline: versus pure_llm (54 discordant pairs in favor of the system, 1 against; $p = 3.1 \times 10^{-15}$), versus naive_rag (60 versus 3; $p = 9.0 \times 10^{-15}$), and versus lightrag (21 versus 1; $p = 1.1 \times 10^{-5}$).

The accuracy gradient full_system > lightrag > pure_llm > naive_rag was stable across all five temperature conditions on this set. The accuracy gap between full_system and the best LLM baseline (lightrag) was 0.170, indicating that the deterministic knowledge engineering contributed more than retrieval augmentation alone. In this 10-token setting naive_rag scored below pure_llm, but that ordering does not survive an adequate output budget (Section 3.6), so we do not interpret it as a property of keyword retrieval.

In the per-subset analysis, the procedurally frozen blind subset ($n = 40$, SHA-locked) reached 0.975 accuracy (39 of 40; Clopper–Pearson CI 0.868–0.999); the single miss is flurbiprofen $\times$ warfarin, discussed in Section 3.5. The FDA/Stockley’s/BNF mechanism

subset (n = 20) reached 1.000. The guideline-sourced subset (n = 58) reached 0.828 (48/58). Eleven exact-scale disagreements remain in total: ten in the guideline subset and the flurbiprofen pair in the frozen subset. The guideline disagreements concern judgment boundaries, including QT risk with low-absorption fluoroquinolones, CYP3A4 inhibition with low-absorption macrolides or azoles, weak-corticosteroid potency granularity, and anticholinergic overlap with tricyclic antidepressants. Each disagreement carries a documented pharmacological rationale (Supplementary Table S1).

3.4 Registry-scale divergence between systemic-route severity and absorption-aware assessment

The largest result in this study does not depend on the gold labels, because both sides of the comparison are pre-specified: DDInter supplies a systemic-route severity rating, and the cascade supplies an absorption-aware grade. We therefore compared every ophthalmic–systemic pair in the 113×232 registries for which DDInter 2.0 provides a severity record — 1,933 pairs. Of these, 1,455 pairs (75.3%) were graded lower by the system, including 352 pairs downgraded from high (37 to medium, 315 to low; 18.2% of all pairs) and 1,103 from medium to low; 136 pairs (7.0%) were upgraded. Of all downgrades, 80.5% (1,171 of 1,455) were attributable to low systemic absorption. In other words, three of every four pairs that carry a severity rating would be presented to a prescriber differently under an absorption-first assessment, and the majority of that shift follows from one pharmacokinetic mechanism rather than from many independent judgements.

Two caveats bound this measurement. First, it quantifies divergence, not error: the analysis shows that the two perspectives rarely coincide for the topical route, not that the

database ratings are wrong. Second, the absorption tiers that drive the downgrades were assigned by the study team from pharmacokinetic sources and regulatory labels (Section 2.2; Supplementary Table S5), so the measurement inherits that curation and should be read as a property of an explicitly stated absorption model rather than as an independent pharmacokinetic measurement.

3.5 Semi-independent audit: DDInter-derived set (92 cases)

On the 92-case audit set, full_system reached 0.674 exact accuracy (62/92; Wilson CI 0.573–0.761) and kappa 0.428 against the absorption-corrected gold (specificity 0.955). Only part of that set is independent of the system’s own absorption assumption: restricting the analysis to the 62 cases kept at the third-party DDInter severity gives 34 of 62 (0.548), which is the figure free of the system’s assumption and should be read as the audit-set headline rather than 0.674. Against the uncorrected DDInter severity labels, agreement was 0.402 (37/92). Stratified: among the 30 absorption-corrected cases (all corrected to low), the system agreed on 28 (93%); among the 62 kept cases, it agreed on 34 (55%) (Table 5). Baselines on the corrected gold: pure_llm 0.630, lightrag 0.591, naive_rag 0.535. McNemar tests on the audit set were significant for full_system versus lightrag (8 versus 1; $p = 0.039$), but not versus naive_rag (27 versus 17; $p = 0.169$) or pure_llm (18 versus 15; $p = 0.728$). We also report, symmetrically, that on this set the two ungrounded baselines exceed the system on high-risk sensitivity (pure_llm and naive_rag 0.848 each, lightrag 0.264, versus 0.320 for the cascade), so the cascade’s advantage here is in specificity and exact accuracy, not in sensitivity.

High-risk sensitivity on the audit set was 0.320 (8 of 25 gold-high cases; CI 0.172–0.516), the lowest of the four methods (pure_llm and naive_rag 0.848, lightrag 0.264). This is the

single most important limitation of the current system and is reported deliberately. The 30 disagreements with the audit gold were directionally consistent with the absorption-first design: 17 gold-high pairs were downgraded (8 to low, 9 to medium), 9 gold-medium pairs to low, 2 gold-medium pairs upgraded to high, and 1 gold-low pair each to medium and high (Supplementary Table S2). The four over-alerts (pairs graded high or medium by the system against a lower gold) were apraclonidine × midazolam, apraclonidine × looperamide, tacrolimus × ibuprofen, and bromfenac × enalapril.

What the system could miss. The 17 downgraded gold-high pairs include six pairs where systemic-route reasoning is weak on ophthalmic grounds (for example aflibercept × leflunomide, with no plausible topical systemic pathway). The remaining downgrades are clinically sensitive: betaxolol × verapamil and betaxolol × diltiazem (selective beta-blockade plus calcium-channel blockade), timolol × epinephrine, scopolamine or homatropine × topiramate (anticholinergic burden), and flurbiprofen × methotrexate. That last pair deserves comment because it illustrates how the absorption model interacts with the third-party labels. An earlier analysis of this registry tiered flurbiprofen high and therefore traced the divergence to a matrix cell graded below its gold. Reviewing the assignment showed it to be a data-entry error: flurbiprofen sodium ophthalmic is formulated at 0.03%, the lowest concentration of any ophthalmic NSAID in the registry, the other four ophthalmic NSAIDs are all tiered low, and the curation file recorded low for the same two evidence sources. Correcting the tier to low (Section 4.4) moves this pair and three related anticoagulant pairs further from the DDInter-derived gold rather than closer to it, because the systemic-route label is high while a 0.03% topical formulation cannot plausibly reach systemic exposure. We report the corrected direction

rather than the original one: the divergence here is between a systemic-route label and a low-absorption ophthalmic agent, not between a matrix cell and its gold. For a screening tool, this false-negative profile is the clinically critical number, and it defines the calibration priorities in Section 4.5.

3.6 Reasoning-model baselines under an adequate output budget

Because the primary comparison used a single non-reasoning model with a 10-token output cap, we repeated the three LLM baselines under conditions that remove that constraint, and report both settings rather than replacing one with the other. The re-run used a reasoning model (glm-5.3-flash), a 3000-token budget, a fixed temperature of 0.0 with five repeated samples per case, and a pre-specified parsing rule (exact match first, then substring matching in the order high > medium > low, else unknown). The system prompt, retrieval settings and user-prompt templates were unchanged, so the two settings differ only in model, output budget and sampling. All 3,150 calls completed with no errors and no unparseable responses.

Accurate label agreement improved. On the 118-case set, pure_llm rose from 0.475 to 0.576 and naive_rag from 0.415 to 0.585; lightrag was unchanged at 0.737. **High-risk sensitivity, however, fell sharply**, and specificity rose to approximately 1.000:

method	Acc3 (10-token)	Acc3 (3000-token)	sens (10-token)	sens (3000-token)	spec (3000-token)
pure_llm	0.475	0.576	0.771	0.250	1.000
naive_rag	0.415	0.585	0.757	0.214	0.989

method	Acc3 (10- token)	Acc3 (3000- token)	sens (10- token)	sens (3000- token)	spec (3000- token)
lightrag	0.737	0.737	0.779	0.536	1.000
full_system	0.907	0.907	0.964	0.964	0.989

The mechanism is a shift in the predicted label distribution rather than an improvement in discrimination. Under the larger budget the baselines converge on a conservative policy: pure_llm assigns high risk to 6% of cases where the gold contains 24%, and on the audit set to 2% where the gold contains 27%. Because both gold sets are dominated by low and medium grades, this conservatism inflates exact accuracy while the baselines identify 6 to 15 of the 28 gold-high cases (the cascade identifies all 28). We therefore read the accuracy increase as an artefact of the decision rule the model adopts when given room to reason, not as evidence that reasoning improves ophthalmic DDI screening; the sensitivity column is the informative one.

Two further results from this re-run bear on the interpretation of the primary comparison. First, the ordering between retrieval methods reverses: under the 10-token cap naive_rag scored below pure_llm (0.415 versus 0.475), and we previously attributed that to retrieval noise, but with an adequate budget naive_rag scores at or above pure_llm on both sets (0.585 versus 0.576; 0.728 versus 0.609). The earlier gap was a property of the constrained setting, and the claim that keyword retrieval injects noise in this domain is withdrawn. Second, on the audit set the adequate-budget naive_rag baseline reaches 0.728 exact accuracy and 0.400 high-risk sensitivity, both above the cascade (0.674 and 0.320); we report this because it is the one comparison in which a baseline outperforms

the system, and it is not explained by the conservatism described above. Its sensitivity is also unstable across the five samples (0.280 to 0.480), whereas the cascade returns the same grade every time.

Sample-to-sample variation was itself informative. At a fixed temperature of 0.0 the baselines still varied by 0.013 to 0.034 in exact accuracy across the five repeats, which is why the reported figures use a majority vote over five samples rather than a single call; the cascade's SD remained 0.000.

3.7 Determinism and temperature sensitivity

Full_system is deterministic: all five temperature conditions returned identical predictions (SD = 0.000 for every metric), a determinism check rather than a robustness claim. The LLM baselines varied across conditions on the L1 set: lightrag accuracy between 0.720 and 0.771 (SD 0.018), pure_llm between 0.449 and 0.517 (SD 0.024), and naive_rag between 0.390 and 0.432 (SD 0.014), reflecting genuine temperature sensitivity in the LLM baselines. The same pattern held on the audit set, where the LLM baselines varied (SD 0.008–0.039) and the deterministic cascade did not. Because the five conditions differ only in temperature, these standard deviations quantify sampling instability rather than seed variance.

3.8 Expert grade review (two ophthalmologists, 12 cases)

Two independent ophthalmology experts reviewed 12 cases under the grade-review protocol of Section 2.5, with the proposed gold and the system grade both visible (Table 6; Supplementary Table S3). Inter-rater agreement was 91.7% (11 of 12; kappa 0.846, weighted kappa 0.875; Figure 3). Expert–system agreement was 41.7% for expert 1 (5 of

12; kappa 0.125, weighted 0.500) and 33.3% for expert 2 (4 of 12; kappa 0.010, weighted 0.461). The disagreement was systematic: expert 1 rated 7 of 12 cases one tier lower than the system and none higher; expert 2 rated 7 lower and 1 higher. Neither expert rated any of the four system-high cases as high (0 of 4). Four cases were fully concordant between both experts and the system (R03 dexamethasone × rifampicin, R07 levofloxacin × sotalol, R10 fluorometholone × prednisone, R11 fluorometholone × glimepiride). The sole inter-expert disagreement was R08 (betaxolol × verapamil), graded medium by expert 1 and high by expert 2. The maximum achievable unweighted kappa given the marginal distributions is 0.625, so the observed near-zero kappas partly reflect margin incompatibility; the weighted kappas (0.500, 0.461) place the divergence at approximately half a tier.

Crucially, the experts disagreed with the gold labels exactly as much as with the system: expert 1 versus gold kappa 0.125 (agreement 41.7%) and expert 2 versus gold kappa 0.010 (agreement 33.3%), identical to the expert–system values because the system matched its gold on all 12 cases. The divergence is therefore not system-specific; it sits between clinical prior and evidence-graded gold. Consistent with this, the experts explicitly noted that they judged from clinical practice without mechanism-level assessment, and that ophthalmic drops are not routinely screened for systemic DDI in their practice. We interpret this as a perspective gap between clinical prior and evidence-based grading, reported transparently; with $n = 12$ and near-zero kappa, the gap cannot be statistically distinguished from chance-level agreement, and the small sample bounds all expert-based conclusions.

3.9 Limitations

First, the L1 gold shares literature sources with the rules (homogeneous construction) and was author-annotated rather than expert-adjudicated; only the 40 frozen blind cases are independent in a procedural sense, and only 20 of these (blind 1) were free of development exposure. Second, 78 of 118 L1 cases overlap the development-phase validation set by case identity (gold independent, cases not entirely novel). Third, the audit gold applies the system's absorption assumption to 30 of 92 cases; against unadjusted DDInter severity, agreement drops to 0.402, and audit-set high-risk sensitivity is 0.320. Fourth, the 12-case expert exercise is a grade review, not a blinded validation of grading accuracy: both the proposed gold and the system grade were visible to the experts, they did not review mechanism evidence, the cases were selected by the study team, and expert 2's ratings were transcribed by the corresponding author from written responses rather than from a scan. Fifth, headline results reflect the final engine on the final (errata-corrected) gold; earlier engine versions scored higher on the same cases (0.992), and the drift is disclosed rather than hidden. Sixth, evidence for ophthalmic-specific DDI outcomes is sparse, so a substantial part of the rule layer is mechanism-derived. These constraints bound the strength of the accuracy claims and are reported to support a fair assessment.

4. Discussion

4.1 Principal findings

The same ophthalmic–systemic pair receives three different risk assignments depending on who is judging, and this study places all three on common ground. Systemic-route

database severity and absorption-aware assessment disagree for 75.3% of the 1,933 registry-matched pairs that carry a DDInter rating, and that divergence is dominated by a single mechanism: low topical systemic absorption accounts for 80.5% of downgrades. Evidence-graded rules and expert judgment also disagree, and not because of the rules — two ophthalmologists agreed closely with each other (kappa 0.846) yet graded most cases one tier lower than both the cascade and its gold, with expert–gold kappas numerically identical to the expert–system kappas (0.125 and 0.010) precisely because the cascade reproduced its gold on all 12 cases. Because both grades were visible to the experts (Section 2.5), this comparison cannot separate the two assessments; what it does establish is that the grades did not persuade them. The results therefore support absorption-first knowledge engineering as a way of making ophthalmic DDI risk explicit and individually auditable, and they locate the largest source of disagreement not in any one tool but in the absence of a shared grading standard for this route.

The quantitative performance results bound that claim rather than carry it. The cascade agreed with its evidence-graded gold on 0.907 of 118 cases (kappa 0.857) and outperformed three LLM baselines (0.415–0.737 accuracy) with significant McNemar tests, zero variance across five temperature conditions, and millisecond inference. On the semi-independent audit set it again led on exact accuracy (0.674 versus 0.535–0.591) and specificity (0.955), but its high-risk sensitivity was the lowest of the four methods (0.320, 8 of 25), and its advantage over pure_llm and naive_rag on that set was not statistically significant. The procedurally frozen blind subset (n = 40) was classified with 39 of 40 correct. Because the L1 gold shares its literature sources with the rules, the 0.907 figure is best read as the internal consistency of an explicitly evidence-graded encoding; the audit-

set sensitivity is the number that bounds what the system can claim as a safety screen today.

4.2 What we add relative to prior work

Against general-purpose DDI databases. Existing resources such as DDInter provide comprehensive pair-level severity ratings derived from systemic administration [16] but do not encode the decisive ophthalmic modifier, the magnitude of systemic absorption from topical administration, nor dose-route corrections. Our system makes this ophthalmic pharmacokinetic factor an explicit, first-class modelling component [32]. A systemic-absorption tier is assigned to every ophthalmic agent (six values in the registry, four of which the hard constraint forces to low); an absorption-scaling step forces low-absorption pairs to low risk within the matrix layer unless a mechanism-based upgrade applies; and a hard-constraint layer encodes the absorption-first pharmacology, with the final grade taken as the maximum of the rule and matrix layers. **That maximum is the honest limitation of the design.** The rule layer is not absorption-gated, so 202 pairs with a low-or-below tier receive medium or high grades through rules that require no absorption flag — all 95 high grades among them from a single rule, `nsaid_antihypertensive_antagonism`. We could have gated that rule, as the four class-matrix upgrades are gated, and we measured what that would do: gating it lowers agreement with the L1 gold from 0.907 to 0.890, and gating all four NSAID rules lowers it to 0.847. We therefore report the behaviour rather than re-engineer it to fit the gold. That the gold prefers an ungated rule is itself informative: the guideline-sourced subset was labelled from sources that reason about systemic exposure, so part of the gold encodes the same systemic-route assumption the paper argues against. Whether to gate

these rules is a calibration decision that should follow outcome-linked evidence, not gold agreement, and Section 4.5 lists it first. In the 1,933-pair DDInter audit (Section 3.4), 75.3% of matched pairs were graded lower by the system, including 352 pairs downgraded from high (18.2% of all pairs), with 80.5% of downgrades attributable to low systemic absorption. The methodological implication is the one worth stating plainly: a severity rating exported from a systemic-route database is not a valid ophthalmic ground truth, and any evaluation of an ophthalmic DDI tool that treats it as one measures the distance between two assumptions rather than the tool’s accuracy. That is why this study reports agreement with unadjusted systemic-route labels (0.210 and 0.443 on the development and external sets, Section 3.2) as a quantification of the adjustment itself rather than as a performance figure.

Against LLM and retrieval-augmented approaches. LLM screening is non-deterministic and difficult to audit, and retrieval-augmented variants inherit the biases of their corpora [33,34]. Machine-learning predictors built on molecular structure [35] or knowledge-graph representations [36] instead target systemic-route interactions and likewise omit route-specific absorption. In our evaluation the graph-context baseline (named lightrag; Section 2.4) captured only part of the gain even though all LLM baselines received an identical system prompt containing the key absorption facts, and under an adequate output budget the baselines converged on a conservative policy that assigns high risk to 6% of cases where the gold contains 24% (Section 3.6). What separates the cascade from the baselines is therefore not label agreement but the sensitivity-specificity profile and the reproducibility of the decision. Our system is deterministic by construction — it makes no model call, so its reproducibility is a design property rather than a robustness result —

and it is auditable (every decision returns matched rule IDs, decision path, and evidence citations) and fast (about 2 ms per pair on a single CPU), while retaining the full mechanism vocabulary of the knowledge base.

Systematic coverage of the overlooked space. Clinical recall of ophthalmic–systemic DDI concentrates on a small set of well-known agents, most prominently non-selective beta-blockers. Beyond this class, interactions are rarely mentioned in routine practice and are easily overlooked [37]. Our system screens all 113 ophthalmic agents against 232 systemic drugs exhaustively. It surfaces mechanism-based signals that clinicians may not spontaneously consider, including anticholinergic accumulation with tricyclic antidepressants, additive carbonic-anhydrase inhibition, CYP3A4-mediated exposure changes, and weak-corticosteroid potency boundaries. Each signal is traceable to verified evidence.

Validation discipline. Beyond absolute performance, we contribute a four-phase evaluation protocol with explicit disclosure of its own weaknesses. The protocol includes a procedurally frozen blind subset (SHA-locked), a third-party audit component (DDInter severity levels) with absorption-correction strata reported separately, a five-temperature protocol separating deterministic and stochastic methods, and full reporting of all four datasets including the unadjusted-gold reference sets. Homogeneous-construction, overlap, and development-exposure limitations are reported transparently. These elements are rarely reported together in clinical NLP or CDSS studies [38,39].

4.3 The expert–clinical gap: a prior-versus-evidence divergence

Two independent ophthalmologists agreed closely with each other (kappa 0.846) but rated most cases one tier lower than both the system and its gold. Because the expert–gold and expert–system kappas are identical (0.125 and 0.010), the divergence is not a system artifact; it reflects a systematic difference between clinical prior and evidence-graded gold. Three observations support this reading. First, clinically serious events from ophthalmic–systemic DDI are rare, and expert judgments rationally gravitate toward lower risk tiers; consistent with this, neither expert rated any of the four system-high cases as high. Second, the few interactions clinicians do recall are dominated by non-selective beta-blockers, which are prominently discussed in prescribing guidance; the systematic coverage described above is therefore most valuable precisely where clinical recall is weakest. Third, reports of serious adverse events attributable to ophthalmic–systemic interactions are rare in the PubMed literature, although pharmacovigilance analyses of ophthalmic agents do identify systemic signals [29,40]. High alert-override rates in computerized order entry [24] further suggest that a screening tool whose outputs sit one tier above clinical intuition will be treated as noise unless its mechanism-based justifications are presented transparently; our traceable decision paths are designed to do exactly that. Whether the gold, the system, or the clinical prior is the better predictor of patient-relevant outcomes cannot be resolved without outcome-linked data; the honest position is that all three are hypotheses awaiting prospective calibration.

4.4 Limitations

Absorption-tier assignment. The tiers are the most load-bearing variable in the cascade and the least independently documented. They are a study-team assignment (Section 2.2)

with no operational measurement behind every agent, and one defect has been corrected during revision and one remains open. **Corrected:** flurbiprofen was tiered high in the registry while the curation file recorded low for the same evidence sources; at 0.03% it is the lowest-concentration ophthalmic NSAID in the registry and the other four are all tiered low, so the high tier was a data-entry error. It is now tiered low, and the registry-scale analysis in Section 3.4 reflects the corrected value. **Open:** nine of the 113 agents in the “ophthalmic topical” registry are not topical (six intravitreal anti-VEGF agents, intraocular acetylcholine, and two surgical viscoelastics), including aflibercept, which is the example given in Section 3.5 of a pair with no plausible topical systemic pathway. Because the tier drives both the class-matrix forcing and the L3 constraint, the registry-scale divergence reported in Section 3.4 should be read as a property of this stated absorption model rather than as an independent pharmacokinetic measurement; Section 3.4 gives the tier cross-tabulation so the magnitude can be judged. Correcting a second class of assignments — the tiering of non-topical agents — remains future work, and the scenario analysis deposited alongside this manuscript quantifies how the registry-scale figures move under each candidate reassignment.

Bounded clinical impact. The scarcity of direct evidence plausibly reflects that ophthalmic absorption rarely alters systemic drug handling in a clinically detectable way. PubMed yielded few dedicated ophthalmic–systemic interaction studies or case series. The evidence base is populated largely by pharmacokinetic first principles, class-level pharmacology, and extrapolation from systemic data (1,307 curated chunks, but few ophthalmic-specific outcome studies). Ocular drug transfer from systemic administration is likewise complex and incompletely characterized [28]. If the true risk gradient clusters

at the low end for most ophthalmic agents, an evidence-graded system may systematically over-grade relative to clinical practice. Conversely, the audit-set sensitivity of 0.320 shows the system can also under-grade relative to an absorption-corrected gold; both directions bound the current clinical utility. At a realistic 1% prevalence of clinically significant ophthalmic–systemic interactions, the audit operating point (sensitivity 0.320, specificity 0.955) would yield a positive predictive value of approximately 9%, so the system’s value lies in systematic coverage and traceability rather than in screening yield.

Validation-related limitations. The L1 gold labels were constructed from the same literature sources that informed the rules (homogeneous construction); the set is a literature-derived, author-annotated proxy rather than an expert-adjudicated gold. Only the 40 procedurally frozen blind cases are independent in the procedural sense, with 20 of them free of development exposure. In addition, 78 of 118 L1 cases overlap the development-phase internal validation set by case identity. No real human expert participated in gold construction. The two-expert grade review is the only human channel. It cannot serve as an independent test of the grading: both the proposed gold and the system grade were visible to the reviewers, and the system reproduced its gold on all 12 cases, so expert-versus-gold and expert-versus-system agreement are necessarily the same comparison and the observed one-tier gap cannot be attributed to either source. What the exercise does show is that two ophthalmologists, given no mechanism evidence, did not endorse the assigned grades, and that this divergence is stable between them; with $n = 12$ and near-chance kappa it cannot be distinguished from chance-level agreement. One expert’s ratings were transcribed by the study team from written responses. Headline

results reflect the final engine and the final, errata-corrected gold; the earlier 0.992 snapshot is disclosed in the process audit. These constraints bound the strength of the accuracy claims.

4.5 Future directions

Four directions follow directly, with the first taking precedence. **Registry correction:** flurbiprofen has been corrected in this revision; the remaining work is to re-assign the tiers with a documented source and threshold for every agent and to reclassify or exclude the nine non-topical agents, then re-run the registry-scale analysis of Section 3.4. The tier table deposited with this work (Supplementary Table S5) and the scenario analysis in the deposit (audits/tier_alternative_scenarios.json) are the starting points. The same item covers the rule-layer question raised in Section 4.2: whether nsaid_antihypertensive_antagonism and the other NSAID rules should be gated on absorption, as the class-matrix upgrades are. We left them ungated because gating reduces agreement with the present gold, but that is a reason to revisit the gold and the calibration jointly rather than a reason to leave the design as it stands. Second, clinical calibration: a larger expert panel reviewing mechanism-level evidence, or outcome-linked retrospective data, could determine whether a systematic one-tier offset should be applied to specific absorption tiers or drug classes, converting the perspective gap into a calibrated prior. The audit-set false-negative pairs (notably flurbiprofen $\times$ methotrexate and betaxolol $\times$ calcium-channel blockers) are the first calibration targets. Third, false-negative surveillance: targeted pharmacovigilance mining (FAERS-style) and prospective monitoring of high-absorption agents would test the mechanism-derived rules against real-world outcomes and populate the sparse evidence base. Fourth, coverage and fail-

safe behaviour: extension of the registry to additional ophthalmic and systemic agents, and a user-visible unassessed status for out-of-coverage pairs rather than a scored default, are planned. Fifth, scope extension: the class-matrix architecture transfers naturally to other specialty–systemic interaction domains (dermatologic, inhaled, and intranasal routes) with the same absorption-first design.

5. Conclusion

We developed OphDDI-CDS, an absorption-first deterministic cascade for ophthalmic–systemic DDI screening (40 mechanism rules; an absorption-scaled class matrix; a hard-constraint layer encoding topical systemic absorption), and used it to compare three ways of assigning risk to the same drug pairs. Against DDInter severity, absorption-aware assessment graded 75.3% of all 1,933 registry-matched pairs lower, with 80.5% of that shift attributable to low topical systemic absorption; a two-expert grade review likewise found systematic disagreement with the assigned grades, independently of the system only in the sense that neither the gold nor the system grade persuaded the reviewers (Section 2.5 limits what this comparison can establish). The cascade itself agreed with its evidence-graded gold on 0.907 of 118 cases (kappa 0.857), with complete output traceability, outperformed three LLM baselines on exact accuracy, and on a semi-independent audit set reached 0.674 exact accuracy — falling to 0.548 on the subset free of its own absorption assumption — but the lowest high-risk sensitivity of the four methods (0.320). That sensitivity, together with a literature-derived author-annotated gold, bounds what the system can claim today. The practical conclusion is that ophthalmic DDI risk currently lacks a shared grading standard: database severity, evidence-graded rules, and clinical prior disagree systematically, and evaluation and calibration in this domain

should measure that disagreement rather than assume any one of the three is correct. Prospective, outcome-linked calibration is the next step, and the false-negative pairs identified here are its first targets.

Keywords

ophthalmic drug-drug interaction; clinical decision support; deterministic rule engine; systemic absorption; explainable artificial intelligence; medication safety

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki and was reviewed and approved by the Ethics Committee of the First Affiliated Hospital of Xinjiang Medical University (approval no. K202509-29; reviewed by expedited review). The approval covers the expert-opinion component of this work. No patient data, human tissue, biological specimens, or clinical interventions were involved: the only human involvement was an expert-opinion exercise in which two ophthalmologists independently rated 12 hypothetical drug-pair vignettes constructed by the study team, no identifiable information was collected, and both experts provided informed consent for their anonymised ratings to be used in research and publication. The ethics approval document is provided as supplementary material. The system is a research decision-support tool and is not intended to replace clinical judgment (Section 2.6).

Consent for publication

Not applicable. This study did not involve data from any individual person.

Availability of data and materials

All datasets supporting the conclusions of this article are deposited in a public repository and are also included as additional files.

Repository. <https://github.com/wangguoqing932/ophddi-cds> (release tag v1.0-submission; accessible without an account). The deposit contains the entity registries (113 ophthalmic, 232 systemic agents) and the 1,307-chunk evidence base; all gold files with case-level provenance (L1 118 cases, DDInter audit 92 cases, development 200 cases, external 70 cases, plus the two development-gate sets); the 12-case expert ratings together with the review instrument and scanned questionnaires; the complete prediction files (9,600 rows, covering 4 methods x 5 temperature conditions x 4 datasets) and the evaluation and statistics scripts; the exact prompt templates; the DDInter severity tables used for the registry-scale audit; a per-agent absorption tier table with a source column for each agent; the path-attribution audit reporting which layer produced each grade; and SHA-256 fingerprints for the engine, rules and gold versions. The deterministic cascade requires no API key or network access, so the pipeline can be run end to end from the deposit; re-running the LLM baselines requires credentials for the model API.

Software. Project name: OphDDI-CDS (Python package `ophthalmic_ddi_cds_agent`). Project home page: <https://github.com/wangguoqing932/ophddi-cds>. Archived version: the release corresponding to this manuscript is tagged v1.0-submission in that repository; a versioned DOI will be registered on acceptance. Operating system(s): platform independent. Programming language: Python 3.11. Other requirements: Python 3.10 or later, with dependencies pinned in `pyproject.toml`. Licence: Apache-2.0 (code) and CC BY 4.0 (data). Any restrictions to use by non-academics: none.

The repository contains the production screening engine, the curated knowledge base, the rule-citation verification tooling and the evaluation scripts, together with a synthetic demonstration dataset. The deterministic cascade described in this paper makes no LLM calls and requires no credential or network access, so the pipeline can be executed end to end with a single command (python scripts/demo_pipeline.py), which reproduces the expected risk level for 25 bundled mock pairs covering all three decision paths and the unknown-drug fail-safe behaviour. A test suite (pytest tests/, 34 passing, 1 skipped) verifies the registry, rules and citation-verification layers.

Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Authors' contributions

Wang G: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing - original draft. Hu L: Clinical input, Data interpretation, Writing - review & editing. Wang S: Clinical input, Data interpretation, Writing - review & editing. Yi X: Conceptualization, Supervision, Project administration, Funding acquisition, Resources, Writing - review & editing. All authors contributed to the article and approved the submitted version.

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Not applicable.

Generative AI

Generative AI tools were used during this research and manuscript preparation, with full author review and responsibility. DeepSeek-chat (deepseek.ai) was used for LLM-baseline evaluations, for literature-assisted drafting, and as an assistive tool in knowledge-engineering tasks, including registry curation, evidence grading, and gold-label annotation; Zhipu GLM-4.6V was used for visual verification of scanned expert questionnaires and figures. Figure 1 (system architecture schematic) was created with an AI image-generation tool from a prompt defined by the authors (prompt archived in the supplementary material, figure_prompts.md); all quantitative figures (Figures 2–5) were generated directly from system outputs with Matplotlib. All AI-assisted content was reviewed and validated by the authors, who take full responsibility for the content of this manuscript. The authors disclose that no human expert participated in gold-label construction; the 12-case expert assessment is the sole human validation channel.

Table 1. Registry summary of the knowledge base (113 ophthalmic agents, 232 systemic agents).

entity_type	count	with_flags	with_drugbank_ids
ophthalmic_drugs	113	113	103
systemic_drugs	232	232	218

Table 2. Rule inventory: 40 mechanism rules plus 3 patient-factor policies (43 policy entries) with severity, risk type, flags, and match semantics.

rule_id	severity	risk_type	entity_a_flags	entity_b_flags	match	
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additive_beta_blockade	high	pharmacodynamic_cardiovascular	beta_blocker	systemic_absorption_high	beta_blocker_systemic	all
beta_blocker_ccb_additive	high	pharmacodynamic_cardiovascular	beta_blocker	systemic_absorption_high	l_type_calcium_channel_blocker	all
beta_blocker_digoxin_additive	high	pharmacodynamic_cardiovascular	beta_blocker	systemic_absorption_high	na_k_atpase_inhibitor	all
additive_alpha2_cns_depression	high	pharmacodynamic_cns	alpha2_agonist	cns_depressant	any	
additive_anticholinergic_effects	medium	pharmacodynamic_anticholinergic	anticholinergic	anticholinergic_systemic	any	
additive_corticosteroid_iop	medium	pharmacodynamic_ocular_hypertension	corticosteroid	corticosteroid_systemic	any	
additive_nsaid_bleeding	medium	pharmacodynamic_hemorrhagic	nsaid	anticoagulant	any	
sympathomimetic_alpha_interaction	medium	pharmacodynamic_cardiovascular	sympathomimetic	antihypertensive	any	
qt_prolongation_risk	low	pharmacodynamic_cardiotoxic	fluoroquinolone	qt_prolonging	any	
pga_antiplatelet_bleeding	low	pharmacodynamic_hemorrhagic	prostaglandin_analog	antiplatelet	any	
cai_diuretic_metabolic_acidosis	medium	pharmacokinetic_metabolic	cai_topical	diuretic	any	
cai_topiramate_acidosis	medium	pharmacokinetic_metabolic	cai_topical	cai_systemic	any	
dexamethasone_cyp3a4_induction	medium	pharmacokinetic_cyp3a4	corticosteroid	systemic_absorption_high	cyp3a4_substrate	all
topical_corticosteroid_cyp3a4_induction	low	pharmacokinetic_cyp3a4	corticosteroid	cyp3a4_substrate	any	
systemic_absorption_high_risk	low	systemic_absorption	systemic_absorption_high	beta_blocker_systemic	any	
systemic_absorption_moderate_risk	low	systemic_absorption	systemic_absorption_medium	narrow_therapeutic_index	any	
p_gp_mediated_absorption	low	pharmacokinetic_transporter	p_gp_substrate	p_gp_inhibitor	any	
calcineurin_inhibitor_cyp3a4	low	pharmacokinetic_cyp3a4	calcineurin_inhibitor	cyp3a4_inhibitor	any	
macrolide_cyp3a4_inhibition	low	pharmacokinetic_cyp3a4	macrolide	cyp3a4_substrate	any	
azole_cyp3a4_inhibition	low	pharmacokinetic_cyp3a4	azole	cyp3a4_substrate	any	
sympathomimetic_maoi_hypertension	high	pharmacodynamic_cardiovascular	sympathomimetic	maoi	any	
beta_blocker_beta2_agonist_antagonism	high	pharmacodynamic_respiratory	beta_blocker	beta2_adrenergic_agonist	any	
immunosuppressant_nsaids_nephrotoxicity	medium	pharmacodynamic_renal	calcineurin_inhibitor	cox1_cox2_inhibitor	any	
fluoroquinolone_corticosteroid	low	pharmacodynamic_muscle	fluoroquinolone	corticosteroid_systemic	any	

eroid_tendon		culoskeletal		ystemic		
nsaid_antihypertensive_antagonism	high	pharmacodynamic_cardiovascular	nsaid	antihypertensive	any	
nsaid_diuretic_antagonism	medium	pharmacodynamic_cardiovascular	nsaid	ncc_inhibitor	nkcc2_inhibitor	any
antibiotic_warfarin_enhancement	low	pharmacokinetic_anticoagulation	antibiotic	narrow_therapeutic_index	vitamin_k_epoxide_reductase_inhibitor	all
beta_blocker_cyp2d6_metabolism	medium	pharmacokinetic_cyp2d6	beta_blocker	cyp2d6_inhibitor	any	
nsaid_acei_arb_antagonism	medium	pharmacodynamic_cardiovascular	nsaid	ace_inhibitor	angiotensin_ii_receptor_antagonist	any
corticosteroid_nti_interaction	medium	pharmacodynamic_endocrine	corticosteroid	narrow_therapeutic_index	any	
sympathomimetic_tca_hypertension	high	pharmacodynamic_cardiovascular	sympathomimetic	net_inhibitor	any	
anticholinergic_qt_prolongation	medium	pharmacodynamic_cardiotoxic	anticholinergic	qt_prolonging	any	
moderate_abs_antiplatelet	low	pharmacodynamic_hemorrhagic	systemic_absorption_medium	antiplatelet	any	
moderate_abs_beta_blocker	medium	pharmacodynamic_cardiovascular	systemic_absorption_medium	beta_blocker_systemic	any	
moderate_abs_anticoagulant	low	pharmacodynamic_hemorrhagic	systemic_absorption_medium	anticoagulant	any	
moderate_abs_cyp2d6	low	pharmacokinetic_cyp2d6	systemic_absorption_medium	cyp2d6_substrate	any	
moderate_abs_diuretic	low	pharmacokinetic_metabolic	systemic_absorption_medium	diuretic	any	
alpha2_agonist_antiplatelet	low	pharmacodynamic_hemorrhagic	alpha2_agonist	antiplatelet	any	
low_abs_antiplatelet	low	pharmacodynamic_hemorrhagic	systemic_absorption_low	antiplatelet	any	
low_abs_anticoagulant	low	pharmacodynamic_hemorrhagic	systemic_absorption_low	anticoagulant	any	

929 **Table 3. Four-method performance on the two validation sets (mean over five temperature**
930 **conditions).**

dataset_id	setting	method	n	sensitivity_high	specificity_high	f1_high	accuracy_3level	cohens_kappa
blind_11	10-token	pure_llm	118	0.771	0.542	0.476	0.475	0.256
blind_11	10-token	naive_rag	118	0.757	0.462	0.434	0.415	0.182
blind_11	10-token	lightrag	118	0.779	0.936	0.785	0.737	0.603
blind_11	10-	full_system	118	0.964	0.989	0.964	0.907	0.857

	token							
blind_l1	3000-token	pure_llm	118	0.250	1.000	0.400	0.576	0.333
blind_l1	3000-token	naive_rag	118	0.214	0.989	0.343	0.585	0.341
blind_l1	3000-token	lightrag	118	0.536	1.000	0.698	0.737	0.587
blind_v3_ddinter	10-token	pure_llm	92	0.848	0.704	0.642	0.630	0.396
blind_v3_ddinter	10-token	naive_rag	92	0.848	0.579	0.570	0.535	0.290
blind_v3_ddinter	10-token	lightrag	92	0.264	0.949	0.377	0.591	0.220
blind_v3_ddinter	10-token	full_system	92	0.320	0.955	0.444	0.674	0.428
blind_v3_ddinter	3000-token	pure_llm	92	0.080	1.000	0.148	0.609	0.317
blind_v3_ddinter	3000-token	naive_rag	92	0.400	1.000	0.571	0.728	0.553
blind_v3_ddinter	3000-token	lightrag	92	0.240	1.000	0.387	0.674	0.428

Note: exact accuracy is on the three-level scale; 10-token denotes the primary comparison (non-reasoning model, 10-token cap, one call per case); 3000-token denotes the re-run (reasoning model glm-5.3-flash, 3000-token budget, temperature 0.0, majority vote over five repeated samples). full_system is deterministic and therefore identical under both settings (SD = 0). Sensitivities are computed against each dataset's gold; the 3000-token baselines assign high risk far more rarely, which raises specificity to ~1.000 while depressing sensitivity.

Table 4. Gold-level distribution of the validation sets.

dataset_id	gold_level	count	percentage
blind_l1	high	28	23.7%
blind_l1	medium	51	43.2%
blind_l1	low	39	33.1%
blind_v3_ddinter	high	25	27.2%
blind_v3_ddinter	medium	19	20.7%
blind_v3_ddinter	low	48	52.2%
blind_l1	total	118	100.0%
blind_v3_ddinter	total	92	100.0%
all	total	210	-

Table 5. DDInter audit strata and disagreement composition (92 cases).

stratum	n	system agreement
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All 92 (vs corrected gold)	92	62 (0.674)
All 92 (vs uncorrected DDInter severity)	92	37 (0.402)
Absorption-corrected cases	30	28 (0.933)
Kept cases (third-party-only)	62	34 (0.548)

938

disagreement direction	n
high->low	8
high->medium	9
medium->low	9
medium->high	2
low->medium	1
low->high	1
total disagreements	30

939 Note: values reflect the corrected flurbiprofen absorption tier. The four over-alerts (system grade above gold) are

940 apraclonidine x midazolam, apraclonidine x loperamide, tacrolimus x ibuprofen and bromfenac x enalapril.

941 **Table 6. Expert assessment: per-case ratings (gold, system, expert 1, expert 2).**

case	ophthalmic	systemic	gold	system	expert1	expert2
R01	Ketorolac	Enoxaparin	medium	medium	low	low
R02	Brimonidine	Tramadol	high	high	medium	medium
R03	Dexamethasone	Rifampicin	medium	medium	medium	medium
R04	Atropine	Tramadol	high	high	medium	medium
R05	Scopolamine	Amitriptyline	high	high	medium	medium
R06	Atropine	Oxycodone	high	high	medium	medium
R07	Levofloxacin	Sotalol	low	low	low	low
R08	Betaxolol	Verapamil	medium	medium	medium	high
R09	Ketorolac	Rivaroxaban	medium	medium	low	low
R10	Fluorometholone	Prednisone	low	low	low	low
R11	Fluorometholone	Glimepiride	low	low	low	low
R12	Ketorolac	Warfarin	medium	medium	low	low

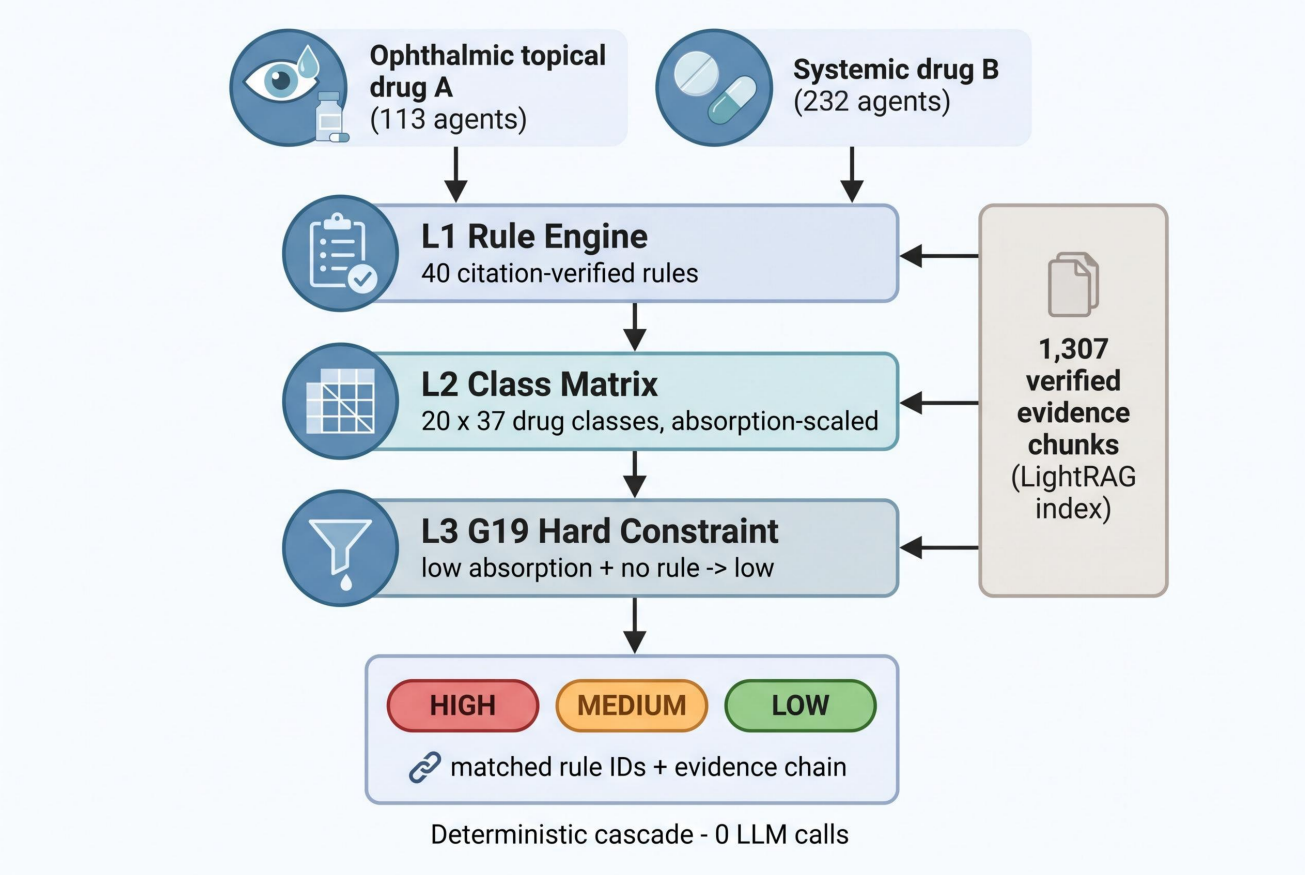


Figure 1. System architecture of the ophthalmic drug–drug interaction clinical decision support system.**

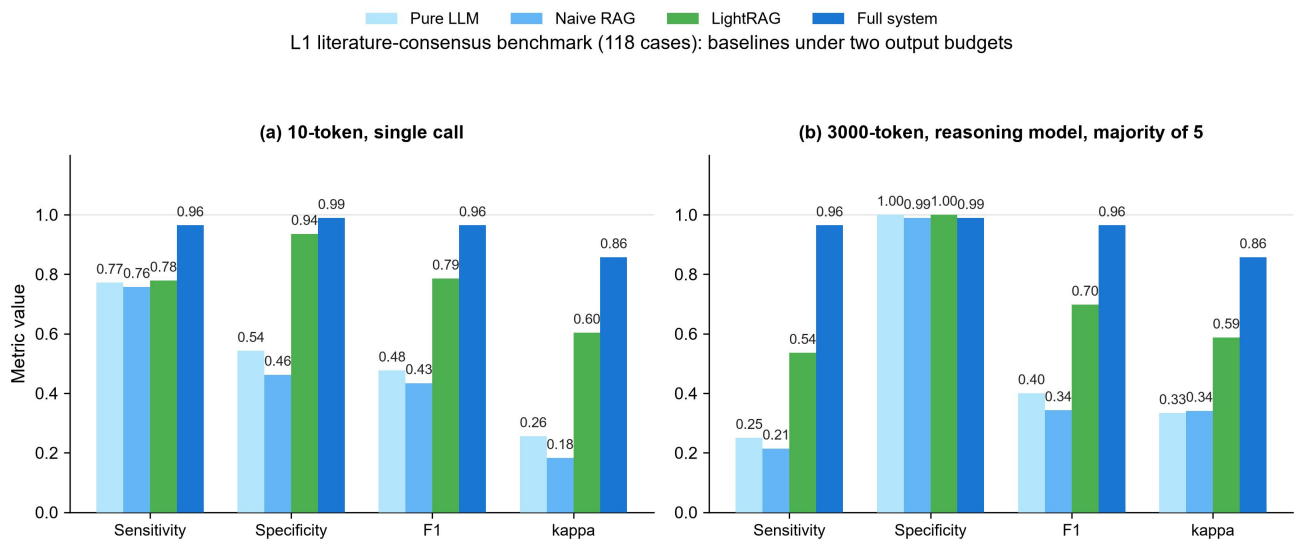


Figure 2. Performance comparison across four methods and two validation sets.**

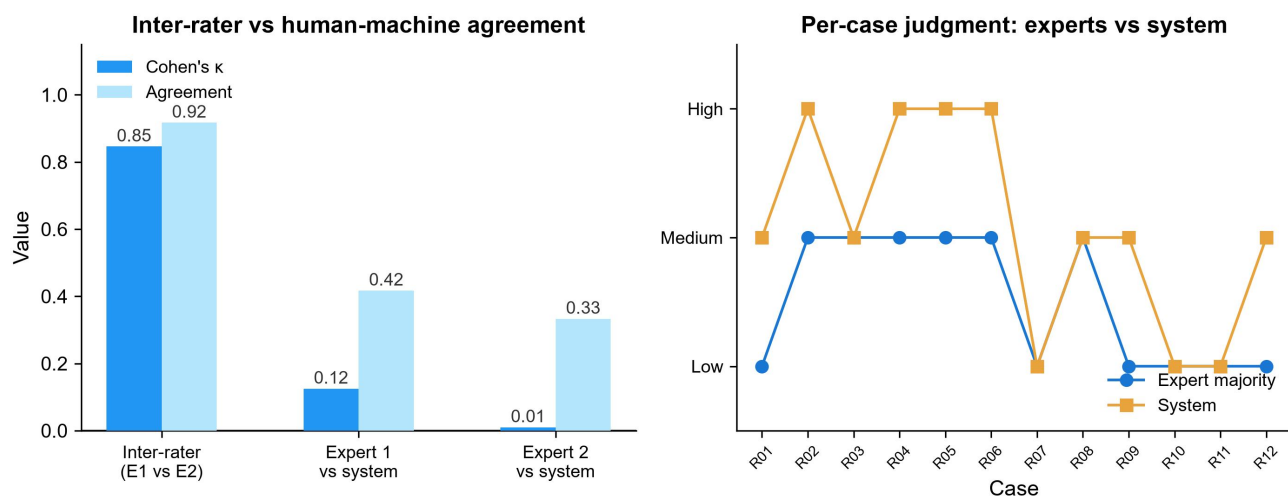


Figure 3. Expert-system agreement and inter-rater reliability (two clinical experts, 12 cases).**

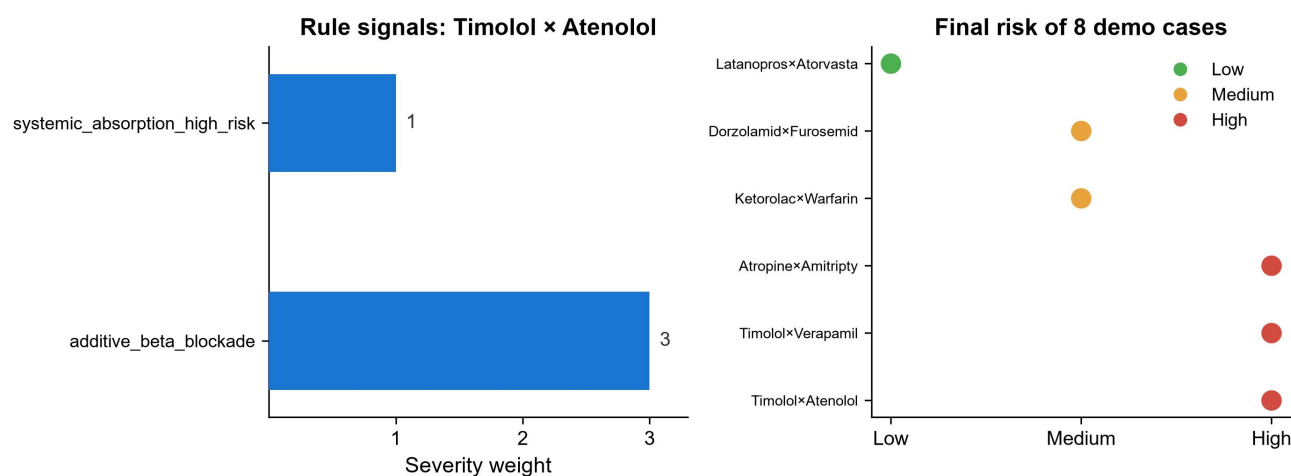


Figure 4. Worked demo case: ophthalmic x systemic risk evaluation (Timolol x Atenolol).**

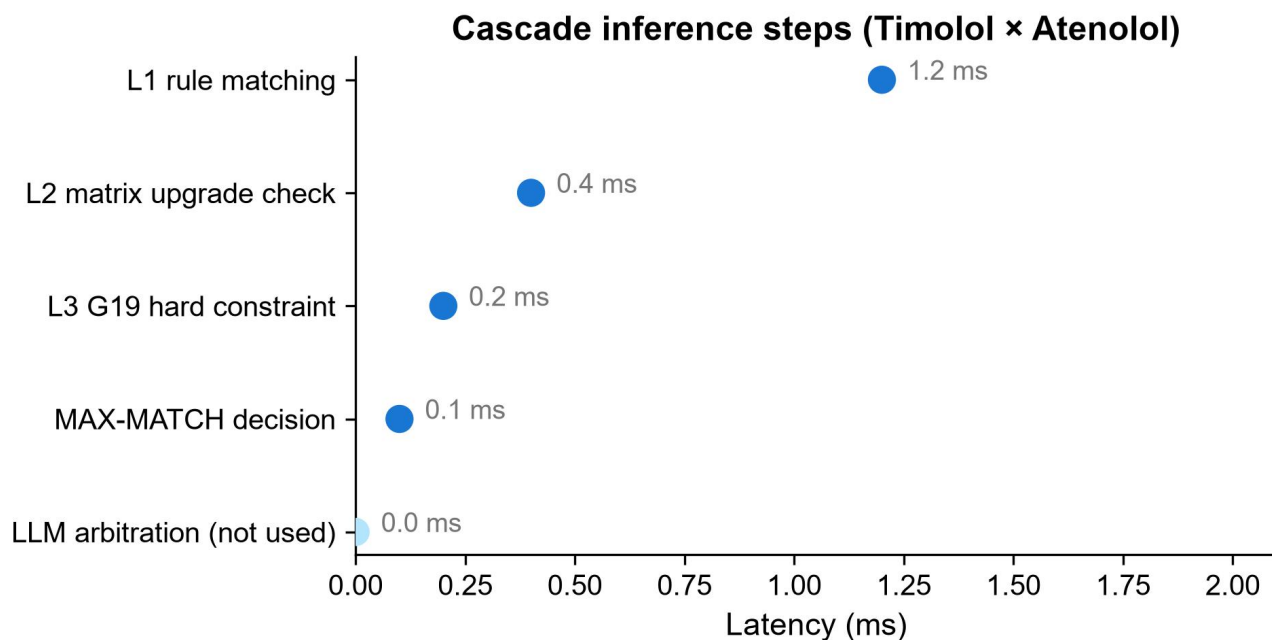


Figure 5. Step-by-step reasoning trace across the cascade layers (Timolol × Atenolol).^{}**

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